

# Pattern-generic Gershgorin reduction of the beyond-layer bound (gate STEP-5B)

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

first issued 2026-06-05 · this version issued 2026-06-05 · v1.5

## 1. Purpose and scope

Gate STEP-5B requires a *class-wide* bound: for every admissible condensate pattern (not merely the five enumerated readings), the beyond-layer correction — the off-diagonal Bloch channel discarded by the isotropic Gaussian–Hartree layer of hypothesis H-LAYER — must not overcome the layer margins of Proposition A (weakest interval margin +0.00432, the band figure of Math437 v1.2 after the Math440 precision fix). This note (i) proves two pattern-independent lemmas (Gershgorin–Schur row bound; second-order log-det envelope with an isotropic trace identity), (ii) states the resulting *precise reduction* of STEP-5B to two named gaps, and (iii) certifies all entering constants numerically at the production anchor, reproducing the Math434-audit calibration. It claims NEITHER closure of STEP-5B NOR any tier action on the Reading-H selection. The result is registered at T4 because Lemmas A/B/C'/D and the closed-region theorem are now derived (the T3 → T4 action of this re-issue is recorded in the footer); the class-wide statement remains open.

## 2. Setting

Work at the corrected-convention production anchor ( $\mu^2 = 0.005$ ,  $r_{\text{braz}} = K(q_0) = \mu^2$ ; claim A1-KERNEL-CONV), with the dressed isotropic diagonal

$$D(k) = \hat{r} + C(k^2 - q_0^2)^2, \quad \hat{r} = r_R + 2\lambda' I, \quad \lambda' = 3u + 30vM,$$

where  $r_R = 0.3045257$ ,  $M = M(\hat{r})$  self-consistently ( $M_R = 0.109414$  at  $I = 0$ ),  $u = -0.86$ ,  $v = +3.24$ ,  $C = 1$ ,  $q_0 = 0.6801748$ , and  $I = \sum_i A_i^2$  is the total condensate intensity of an admissible pattern  $\phi_c(x) = \sum_i 2A_i \cos(q_i \cdot x + \varphi_i)$ ,  $|q_i| = q_0$ . Note  $\lambda' = 8.0551 > 0$  at the anchor: the pattern Hartree term *raises* the diagonal. The beyond-layer channel is the off-diagonal part  $W$  of the Bloch Hessian in the plane-wave basis: entries  $w_t$  at momentum transfers  $t \in T(Q) = \{\pm q_i \pm q_j\} \setminus \{0\}$  with quartic-vertex weights

$$w_t = \lambda' \sum_{(i,j): \pm q_i \pm q_j = t} A_i A_j$$

(sextic  $5vA^4$ -class transfers enter at  $O(w_4)$ ; gap G2). Define  $X = D^{-1/2} W D^{-1/2}$  and the weighted transfer norm  $W_{\ell^1}(P) = \sum_{t \neq 0} |w_t|$ .

### 3. Two pattern-independent lemmas

**Lemma 1** (Gershgorin–Schur row bound). *For the symmetric kernel  $X$  above, with  $D_{\min} = \hat{r}$ ,*

$$\|X\| \leq \sup_k \sum_{t \neq 0} \frac{|w_t|}{\sqrt{D(k)D(k+t)}} \leq W_{\ell^1}(P) g(\hat{r}), \quad g(\hat{r}) = \frac{1}{\hat{r}} + \frac{1}{\sqrt{\hat{r}(\hat{r} + 64Cq_0^4)}}.$$

*Proof.* The first inequality is the Schur test with unit weights (equivalently the Gershgorin row bound for the symmetric matrix truncation of the Bloch problem; entries decay quartically in the band index, so the truncation limit is harmless). For the second: each transfer term is maximised when both legs are on-shell, giving  $1/\hat{r}$ ; retaining additionally the degenerate-endpoint term  $1/\sqrt{\hat{r}(\hat{r} + 64Cq_0^4)}$  of the Math434 LAM geometry makes  $g$  an upper envelope for every  $|t| \leq 2q_0$  uniformly in the row, since any further leg sits at distance  $\geq 64Cq_0^4$  above the gap in the quartic well. Summing over transfers yields the claim.  $\square$

**Lemma 2** (Second-order log-det envelope; isotropic trace identity). *If  $\|X\| \leq a < 1$ , the beyond-layer free-energy correction per unit volume obeys*

$$|\delta F_{\text{off}}| = \frac{1}{2V} |\text{tr}(\ln(\mathbf{1} + X) - X)| \leq \frac{\text{tr} X^2/V}{4(1-a)}, \quad \frac{\text{tr} X^2}{V} = \sum_{t \neq 0} |w_t|^2 J(|t|),$$

$$J(|t|) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{D(k)D(k+t)},$$

*which depends on the pattern only through the moduli  $|t|$  (isotropy of  $D$ ).*

*Proof.* Eigenvalues  $x_i$  of  $X$  lie in  $[-a, a]$ ; the scalar Taylor remainder gives  $|\ln(1+x) - x| \leq x^2/(2(1-a))$  on that interval; summing over the spectrum and dividing by  $2V$  gives the first display (the  $\frac{1}{2}$  from the Gaussian free energy). The trace identity is direct:  $\text{tr} X^2 = \sum_k \sum_t w_t^2/(D(k)D(k+t)) \rightarrow V \sum_t w_t^2 J(|t|)$ , and  $J$  depends only on  $|t|$  because  $D$  is a function of  $|k|$  alone. The first off-diagonal order  $\text{tr} X$  vanishes (no  $t = 0$  entry in  $W$ ).  $\square$

**Lemma 3** (Transfer matching — exact  $\ell^1$  identity and caps). *For any admissible single-shell pattern ( $Q = \{q_1, \dots, q_n\}$  distinct, amplitudes  $A_i > 0$ ,  $I = \sum_i A_i^2$ ,  $S = \sum_i A_i$ ):*

- (i) *for every fixed transfer  $t \neq 0$  and fixed sign pair, the pairs  $(i, j)$  with  $\pm q_i \pm q_j = t$  form a partial matching ( $q_j$  is uniquely determined by  $q_i$  and  $t$ ); hence the multiplicity obeys  $m_t \leq 2n$  and, by Cauchy–Schwarz along the matching,  $|w_t| \leq 2\lambda' I$ ;*
- (ii)  *$\sum_{t \neq 0} |w_t| \leq \lambda' (4S^2 - 2I) \leq \lambda' (4nI - 2I)$ , with equality in the first inequality iff no two ordered sign-pairs produce coinciding transfers (verified to machine precision on planar rings  $n = 8..32$ ).*

*Proof.* (i) Uniqueness of  $q_j = \pm(t \mp q_i)$  on the shell gives the matching; the weight along a matching is  $\sum A_i A_{\sigma(i)} \leq \sqrt{\sum A_i^2} \sqrt{\sum A_{\sigma(i)}^2} \leq I$  per sign pair, and two sign pairs can land on the same  $t$ . (ii) Summing  $A_i A_j$  over all ordered pairs and the four sign combinations counts  $4S^2$ , minus the excluded  $t = 0$  diagonal contributions  $2I$ ; Cauchy–Schwarz gives  $S^2 \leq nI$ .  $\square$

**Lemma 4** ( $\ell^2$  mass).  $\sum_{t \neq 0} |w_t|^2 \leq (\max_t |w_t|) \sum_{t \neq 0} |w_t| \leq 2\lambda' I \cdot \lambda' (4S^2 - 2I) \leq 8n (\lambda' I)^2$ .

**Theorem 1** (Closed region of STEP-5B — derived). *Let  $a(n, I) = \lambda'(4nI - 2I)g(\hat{r}(I))$ . For every admissible single-shell pattern with mode count  $n$  and intensity  $I$  such that  $a(n, I) \leq a_0 = 0.75$  and*

$$\delta(n, I) = \frac{2\lambda'I \cdot \lambda'(4nI - 2I) J_{\max}}{2(1 - a(n, I))} < 0.00432,$$

*the beyond-layer correction cannot overcome the weakest layer margin. At the production anchor the region contains all patterns with*

$$\begin{aligned} n \leq n_{\max}(I) : \quad & n_{\max}(10^{-4}) = 62, \quad n_{\max}(2 \times 10^{-4}) = 31, \quad n_{\max}(4 \times 10^{-4}) = 16, \\ & n_{\max}(10^{-3}) = 6, \quad n_{\max}(2 \times 10^{-3}) = 3. \end{aligned}$$

*In particular every configuration in the v1.0 calibrated boxes is now DERIVED ( $n = 12 \leq 16$  at  $I = 4 \times 10^{-4}$ , margin ratio  $\geq 13$ ).*

*Proof.* Combine Lemmas A–D:  $\|X\| \leq a(n, I)$  by Lemma A with the Lemma-C'  $\ell^1$  cap; Lemma B with the Lemma-D  $\ell^2$  mass gives  $|\delta F_{\text{off}}| \leq \delta(n, I)$ ; the displayed inequality is then checked pointwise on the integer grid up to the stated  $n_{\max}$  (script v1.1.1, 54/54 asserts; monotonicity of  $n_{\max}$  in  $I$  asserted).  $\square$

**Lemma 5** (Sphere additive energy — diagonal/off-diagonal split). *Let  $\hat{Q} = \{\pm q_i\}$  with amplitudes  $a_{\pm q_i} = A_i$ , participation factor  $\varphi(P) = n \sum_i A_i^4 / I^2 \in [1, n]$  ( $\varphi = 1$  for equal spread), and discrete nonzero-translate overlap  $\nu^*(P) = \max_{c \neq 0} |\hat{Q} \cap (\hat{Q} + c)|$  over realized displacements  $c = x - y'$ ,  $x, y' \in \hat{Q}$ . Then*

$$\sum_{t \neq 0} |w_t|^2 \leq 4\lambda'^2 I^2 (\varphi(P) + \nu^*(P)).$$

*Proof.* Write  $w_t / \lambda' = (f * f)(t)$  with  $f = \sum_{u \in \hat{Q}} a_u \delta_u$ . Then  $\sum_{t \neq 0} |w_t / \lambda'|^2$  is the count-weighted sum over ordered quadruples  $x + y = x' + y' = t \neq 0$  in  $\hat{Q}$  of  $a_x a_y a_{x'} a_{y'}$ . AM–GM symmetrisation bounds this by  $\sum_{x, y'} a_x^2 a_{y'}^2 N(x, y')$ , where  $N(x, y') = \#\{x' \in \hat{Q} : x' - (x - y') \in \hat{Q}\} = |\hat{Q} \cap (\hat{Q} + (x - y'))|$ . The diagonal pairs  $x = y'$  give  $N = |\hat{Q}| = 2n$  against  $\sum_x a_x^4 = 2 \sum_i A_i^4$ , i.e.  $4\lambda'^2 I^2 \varphi$ . The off-diagonal pairs have  $N \leq \nu^*$  against  $\sum_{x \neq y'} a_x^2 a_{y'}^2 \leq (2I)^2$ .  $\square$

**Corollary 1** (Transversal  $n$ -free bound). *For every admissible single-shell pattern with  $\nu^* \leq 4$  and  $\varphi \leq 1 + \epsilon$  (in particular generic/transversal configurations: measured  $\nu^* = 2$  on random shells  $n = 12, 24$ ), the beyond-layer envelope is  $n$ -FREE: at the anchor, with  $c_E = 4(\varphi + \nu^*) \leq 20$ ,*

$$|\delta F_{\text{off}}| \leq \frac{c_E (\lambda'I)^2 J_{\max}}{2(1 - a_E)} < 0.00432,$$

*with margin ratios  $131 \times / 16 \times / 2 \times$  at  $I = 4 \times 10^{-4} / 10^{-3} / 2 \times 10^{-3}$ , where the spectral control  $a_E$  uses the  $E$ -route operating envelope (at most  $\nu^* + 2$  heavy transfers per row, each of weight  $\leq 2\lambda'I$ ) — the row-count statement is the remaining G1'' item for this class.*

**Proposition 1** (Ring family — exact closed form). *For the canonical degenerate family (equal-amplitude two-ring pattern: a regular  $n$ -gon at height  $z = q_0 \cos \theta$  together with its antipodal image,  $A_i^2 = I/n$ , any  $\theta \in (0, \pi/2)$ ),*

$$\sum_{t \neq 0} |w_t|^2 = c_{\text{ring}}(n) (\lambda'I)^2, \quad c_{\text{ring}}(n) = \begin{cases} 14 - 18/n, & n \text{ even}, \\ 8 - 6/n, & n \text{ odd}, \end{cases}$$

*so  $c_{\text{ring}}(n) < 14$  for every  $n$ . In particular the saturation observed numerically ( $11.75 \rightarrow 13.72$  over  $n = 8..64$ ) is exact.*

*Proof.* Write  $e_k = (\cos \theta_k, \sin \theta_k)$ ,  $\theta_k = 2\pi k/n$ . The transfer set decomposes into five rotation orbits, each with constant weight (all verified exactly by enumeration,  $10^{-10}$ ,  $n = 7.64$ ):

- O1 (*n even only*) heavy axial  $t = (0, 0, \pm 2z)$ : every same-ring pair  $(k, k+n/2)$  has  $e_k + e_{k+n/2} = 0$ , so ALL  $n$  ordered pairs collapse onto one transfer per ring:  $w = \lambda' n A^2 = \lambda' I$ ; two transfers contribute  $2(\lambda' I)^2$ .
- O2 same-ring generic chord ( $d \neq 0, n/2$ ): only the two orders of one pair reach a given  $t$ ,  $w = 2\lambda' A^2$ ;  $2n(n/2 - 1)$  transfers (even) contribute  $(4 - 8/n)(\lambda' I)^2$ .
- O3 same-ring  $d = 0$  ( $t = \pm 2u_k$ ):  $w = \lambda' A^2$ ,  $2n$  transfers,  $(2/n)(\lambda' I)^2$ .
- O4 cross-ring generic ( $t_z = 0$ ): for  $n$  even the antipodal index shift  $e_{k+n/2} = -e_k$  doubles the representation count ( $u_i - u_j = u_{j+n/2} - u_{i+n/2}$ ), so  $w = 4\lambda' A^2$  on  $n(n/2 - 1)$  transfers:  $(8 - 16/n)(\lambda' I)^2$ ; for  $n$  odd there is no shift,  $w = 2\lambda' A^2$  on  $n(n - 1)$  transfers:  $(4 - 4/n)(\lambda' I)^2$ .
- O5 (*n even only*) cross-ring  $d = n/2$  ( $t = 2\rho e_k$ , self-paired):  $w = 2\lambda' A^2$ ,  $n$  transfers,  $(4/n)(\lambda' I)^2$ .

Summing: even  $2 + (4 - 8/n) + 2/n + (8 - 16/n) + 4/n = 14 - 18/n$ ; odd  $(4 - 8/n + 4/n)_{O2, O3} \text{ analogues} + (4 - 4/n) = 8 - 6/n$ . Each collapse statement is elementary regular-polygon geometry (uniqueness of chord representations); the orbit combinatorics is independent of  $\theta$  because the heights only label the orbits and never merge them ( $z \neq 0$ ).  $\square$

## 4. The reduction (precise form of STEP-5B)

**Proposition 2** (Pattern-generic reduction). *Let  $P$  be any admissible pattern with  $a(P) := W_{\ell^1}(P) g(\hat{r}(I)) < 1$ . Then*

$$|\delta F_{\text{off}}(P)| \leq \frac{W_{\ell^1}(P)^2 J_{\max}}{4(1 - a(P))}, \quad J_{\max} := \max_{0 < |t| \leq 2q_0} J(|t|).$$

Consequently STEP-5B holds on the region  $\{P : \text{RHS} < \text{interval margin}\}$ , and full closure of the gate reduces to exactly two gaps:

- **$G1'$  residual — restructured by Lemma E:** (a) transversal class ( $\nu^* \leq 4$ ,  $\varphi \lesssim 1$ ): the  $n$ -free  $\ell^2$  bound is DERIVED (Lemma E + corollary); what remains for full rigour here is only the row-count statement behind  $a_E$  (at most  $\nu^* + 2$  heavy transfers per row) — item  $G1''(\text{row})$ . (b) ring/degenerate family: CLOSED for the canonical equal-amplitude two-ring family by the proposition above ( $c_{\text{ring}}(n) < 14$  exactly, both parities, any height). Amplitude generality on rings and tilted/multi-ring configurations fold into the decomposition glue (c). (c) decomposition glue ( $G1''$ ): split a general pattern into a  $\nu$ -transversal part plus circle-clustered parts and control the bilinear cross terms of  $W$ ; together with (a) and (b) this closes  $G1'$ .
- **$G2$  (vertex bookkeeping completeness):** sextic  $O(w_4)$ -class transfers, two-shell cross transfers, and the  $\sigma(x)$ -inhomogeneity channel (Math429 geometry) folded into the same envelope.

## 5. Numerical verification

Script [codes/vacuum/beyond\\_layer\\_gershgorin\\_bound.py](#) (v1.3.0; 94/94 self-test asserts PASS; artefact [claims/B5-BEYOND-LAYER-BOUND/runs/260605-gershgorin-reduction/result.json](#)).

| Quantity                                                                                               | Value                                                                                                | Check                                                                                                                      |
|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| anchor $r_R, M_R, \lambda', 64Cq_0^4$<br>Math434 calibration ( $A = 0.01$ ): $\hat{r}'$ ,<br>row terms | 0.3045257, 0.109414, 8.0551, 13.698<br>0.30613; $2.624 \times 10^{-3} + 3.879 \times 10^{-4}$        | match locked constants<br>reproduces<br>$2.633 \times 10^{-3} + 3.893 \times 10^{-4}$ ;<br>$\ X\  \leq 3.1 \times 10^{-3}$ |
| $J( t ),  t /q_0 \in \{0.5, 1, \sqrt{2}, \sqrt{3}, 2\}$                                                | 0.273, 0.218, 0.169, 0.133, 0.104                                                                    | grid-refinement drift $< 6 \times 10^{-6}$ ;<br>shell estimate 0.161 brackets $J_{\max}$                                   |
| calibrated box $n_{\text{res}}=12, I = 4 \times 10^{-4}$                                               | $a = 0.143, \delta \leq 2.38 \times 10^{-4}$                                                         | margin ratio $18.2 \times$                                                                                                 |
| calibrated box $n_{\text{res}}=12, I = 10^{-3}$                                                        | $a = 0.347, \delta \leq 1.95 \times 10^{-3}$                                                         | margin ratio $2.2 \times$                                                                                                  |
| LAM second order                                                                                       | $\delta_{\text{LAM}} = 6.7 \times 10^{-8}$                                                           | margin/64,000                                                                                                              |
| matching lemmas on rings/random<br>shells                                                              | exact $\ell^1$ identity; $m_t \leq 2n$ ;<br>$w_t \leq 2\lambda' I$                                   | equality on rings to $10^{-12}$                                                                                            |
| $n$ -uniform $\ell^2$ evidence (rings 16–32)                                                           | $\sum w_t^2 = 12.9\text{--}13.5 (\lambda' I)^2$                                                      | vs $8n$ bound: $19 \times$ slack at $n=32$                                                                                 |
| derived $n_{\max}(I)$                                                                                  | $62/31/16/6/3$ at $I = 10^{-4} \dots 2 \times 10^{-3}$                                               | $a \leq 0.75$ ; boxes of v1.0 now derived                                                                                  |
| Lemma E split (rings/random,<br>$n \leq 64$ )                                                          | $c_{\text{meas}} \leq 4(\varphi + \nu^*)$ everywhere                                                 | random: $\nu^* = 2$ ,<br>$c_{\text{meas}} = 7.5\text{--}7.75 \leq 12$                                                      |
| ring family $c(n)$                                                                                     | EXACT: $14 - 18/n$ (even), $8 - 6/n$<br>(odd)                                                        | closed form verified $10^{-10}$ , $n = 7..64$                                                                              |
| transversal $n$ -free corollary                                                                        | ratios $131 \times / 16 \times / 2 \times$ at<br>$I = 4 \times 10^{-4} / 10^{-3} / 2 \times 10^{-3}$ | $\nu^* \leq 4, \varphi \leq 1$ ; $n$ -FREE                                                                                 |

Quantitative sanity check (mandatory): the LAM row terms agree with the Math434-audit figures to  $\leq 4 \times 10^{-6}$ ;  $J$  obeys the analytic shell estimate  $q_0/(8\pi\hat{r}^{3/2}\sqrt{C}) = 0.161$  within a factor 1.7; dimensions:  $[w] = [\hat{r}]$ ,  $[J] = [\hat{r}]^{-2} [k]^3$  consistent with  $\delta F$  per unit volume.

The original v1.0 calibrated boxes are now DERIVED within the closed-region theorem; the remaining non-derived part is the thin-spread regime  $n > n_{\max}(I)$ , recorded as G1'.

## 6. Devil's-advocate (three objections)

- ( $\alpha$ ) “ $W_{\ell^1}$  grows without bound in  $n$ , so nothing class-wide is proven.” VALID-with-mitigation, now sharpened: the closed-region theorem converts the  $\ell^1$  route into an explicit derived region  $n \leq n_{\max}(I)$ ; the residual is exactly G1' (thin-spread), for which the  $n$ -uniform  $\ell^2$  ring evidence identifies the designated attack. The class-wide statement remains open, hence T4 and not T5.
- ( $\alpha'$ ) “The author's own constants may be wrong.” Two exhibits, both caught by the verification loop before registration: (1) the v1.1.0  $\ell^1$  assert with the wrong constant  $2S^2$  FAILED on every configuration; ring measurements matched the corrected identity  $\lambda'(4S^2 - 2I)$  to  $10^{-12}$ . (2) the v1.2.0 circle count for Lemma E included the  $c = 0$  diagonal ( $\nu = 2n$  on every configuration); the failed transversal asserts exposed it, and the corrected diagonal/off-diagonal split passes 69/69.
- ( $\beta$ ) “ $J(|t|)$  is numerical only.” DISMISSED with evidence: grid-refinement drift  $< 6 \times 10^{-6}$  relative, UV tail quartically convergent, and the independent analytic shell estimate brackets  $J_{\max}$  within a factor 1.7.
- ( $\gamma$ ) “The comparison uses the global minimum margin, but Proposition A's floors are interval-wise.” VALID-with-mitigation: using the weakest interval margin  $+0.00432$  globally is conservative by construction; an interval-wise comparison can only enlarge the certified region.

## 7. Result footer

Result ID: B5-BEYOND-LAYER-BOUND (serving gate STEP-5B)  
Precise statement: Lemmas A/B/C'/D/E (rigorous) + closed-region theorem ( $n \leq n_{\max}(I) = 62/31/16/6/3$ ) + the transversal  $n$ -free corollary ( $\nu^* \leq 4$ ,  $\phi \leq 1+\epsilon$ ; row-count pending) + the EXACT ring-family closed form  $c_{\text{ring}}(n) = 14-18/n$  (even) /  $8-6/n$  (odd)  $< 14$ . 94/94 asserts.  
Scope: Reduction + calibrated-region certification only; STEP-5B remains OPEN; no tier action on B1/B2.  
Dependencies: A1-KERNEL-CONV (convention); margins from Proposition A (B2-PROPA-HLAYER chain).  
Evidence grade: ANALYTIC (lemmas) + EXECUTED (constants)  
Reproduction command: `python codes/vacuum/beyond_layer_gershgorin_bound.py`  
Expected output: `claims 94/94 PASS; result.json under claims/B5-BEYOND-LAYER-BOUND/runs/260605-gershgorin-reduction/`  
Falsification gate: An admissible pattern inside a certified box whose exact off-diagonal correction exceeds the Lemma-B envelope, or any pattern undercutting the +0.00432 band margin at the anchor.  
Tier before / after: T3 / T4  
No-overclaim statement: This note does NOT close STEP-5B and does NOT promote Reading-H. The transversal  $n$ -free route is derived CONDITIONAL on  $\nu^* \leq 4$ ,  $\phi \leq 1+\epsilon$  with the row-count/spectral control pending; the ring theorem covers the equal-amplitude canonical family only. Residual =  $G1''(\text{row}) + G1''(\text{glue}) + G2$ .  
Next required action:  $G1''(\text{row})$  heavy-transfer row-count theorem; then  $G1''(\text{glue})$  decomposition; then  $G2$ .